

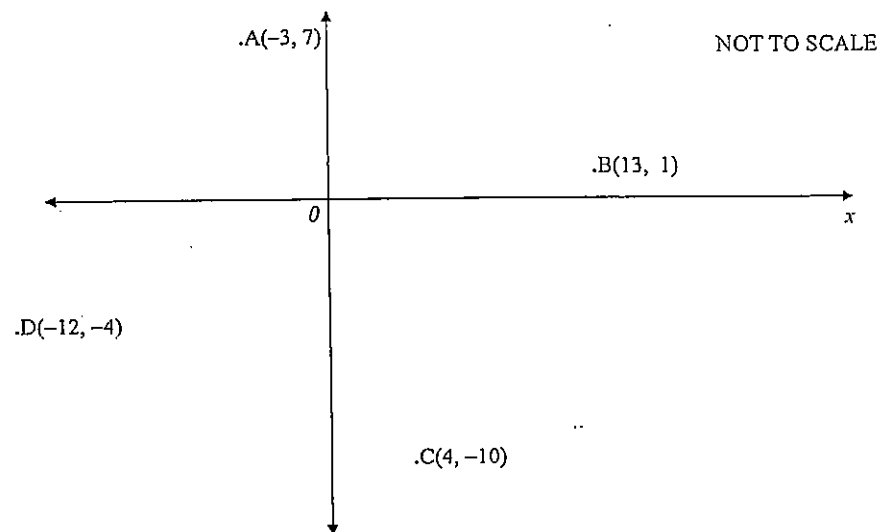
Question 1 (10 marks)

- (a) Simplify the ratio $1.6 : 0.12$ (1)
- (b) Write 0.00618 in scientific notation. (1)
- (c) Rueben scored 43 out of 56 in a Calculus test. Write his result as a percentage correct the nearest percent. (1)
- (d) Convert 270 millimetres to metres. (1)
- (e) Find $\sqrt[3]{710000}$ correct to 2 significant figures. (2)
- (f) If x is an integer, state the set of values satisfying $-2 \leq x+1 < 3$ (2)
- (g) Write $\frac{\sqrt{2}}{2-\sqrt{2}}$ in simplest form and with a rational denominator. (2)

Question 2 (12 marks)

- (a) Expand and simplify $100 - 30(20 - t)$ (1)
- (b) Expand $(x - 3y)^2$ (1)
- (c) Factorise $64 - 8a^3$ (2)
- (d) Solve $\frac{6-x}{2} \leq x$ (2)
- (e) Solve $\left(\frac{1}{27}\right)^{x-2} = (\sqrt{3})^x$ (3)
- (f) Solve $3x - 4 = |2x - 11|$ (3)

Question 3 (12 Marks)

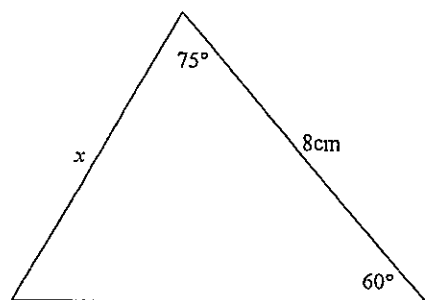


$A(-3, 7)$, $B(13, 1)$, $C(4, -10)$, and $D(-12, -4)$ form the vertices of a quadrilateral $ABCD$.

- (a) Find the length of the diagonal BD . (2)
- (b) Show that AC and BD bisect each other. (2)
- (c) Show that AC is **NOT** perpendicular to BD . (3)
- (d) Explain why $ABCD$ is a parallelogram. (1)
- (e) Show that the equation of BD is $x - 5y - 8 = 0$ (2)
- (f) Find the perpendicular distance from A to BD . (2)

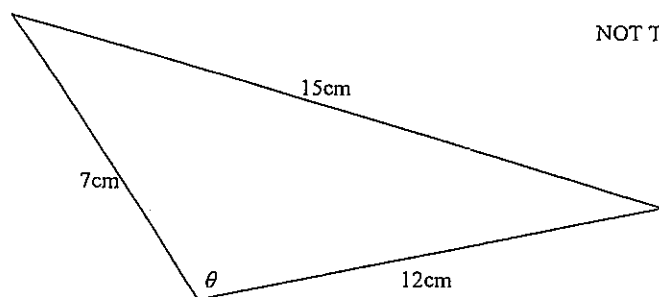
Question 4 (12 marks)

- (a) Find the exact value of x in the triangle below.



NOT TO SCALE

- (b) Find the value of $\cos \theta$ in the diagram, writing your answer as a simplified fraction.



NOT TO SCALE

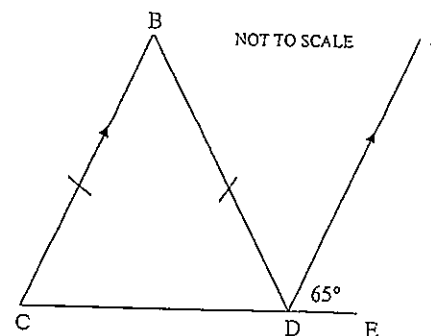
- (c) Given $\cos \alpha = \frac{4}{7}$ and $\sin \alpha < 0$ find the exact value $\tan \alpha$

- (d) Solve $\sin x = \sqrt{3} \cos x$ for $0 \leq x \leq 360^\circ$

- (e) Prove $(\operatorname{cosec}^2 x - 1)(1 - \cos^2 x) = \cos^2 x$

Question 5 (18 marks)

- (a)

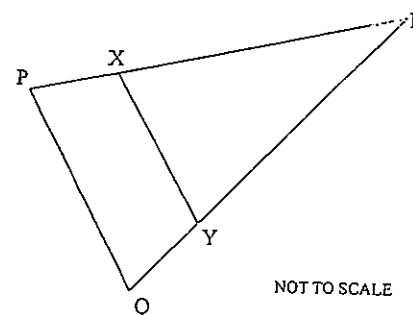


NOT TO SCALE

In the diagram $\angle ADE = 65^\circ$, $BC = BD$ and BC is parallel to AD .

Find, giving reasons, the size of $\angle CBD$.

- (b)



NOT TO SCALE

In the diagram $PR = 36\text{cm}$, $PX = 12\text{cm}$, $QR = 30\text{cm}$ and $QY = 10\text{cm}$.

- (i) Prove $\triangle PRQ \parallel \triangle XRY$. (3)

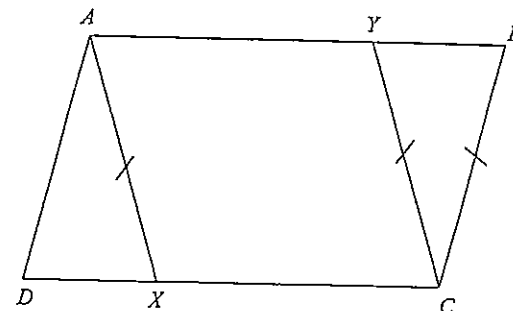
- (ii) Hence, or otherwise, state why $\angle RPQ = \angle RXY$. (1)

- (iii) State why PQ is parallel to XY . (1)

- (iv) If $XY = 16\text{cm}$, find PQ . (2)

- (c) Find the number of sides of a polygon if the sum of the interior angles is 3780° . (2)

- (d) $ABCD$ is a parallelogram. The point X lies on CD , the point Y lies on AB , and $AX = CY = BC$, as shown in the diagram.



NOT TO SCALE

- (i) Explain why $\angle ADX = \angle CBY$ (1)

- (ii) Show that $AD = AX$ (2)

- (iii) Show that triangles ADX and CBY are congruent. (3)

Question 6 (18 marks)

(a) Neatly sketch of each of the following curves on separate sets of axes. Label all critical points.

(i) $y = \sqrt{9 - x^2}$ (3)

(ii) $y = 1 + 2^{-x}$ (3)

(iii) $y = (1 - x)^3$ (3)

(b) State the domain and range of $y = \frac{5}{2 - x}$ (2)

(c)

$$\text{If } f(x) = \begin{cases} 6, & x < -1 \\ x^2 + 6, & -1 \leq x < 2 \\ 3 - x, & x \geq 2 \end{cases}$$

Find the value of $f(-1) + f(-12) - f(5)$ (2)

(d) Show that $g(x) = \frac{3x}{3 + x^2}$ is an odd function (2)

(e) Draw a sketch of the region satisfying each of the following inequalities simultaneously (3)

$$\begin{cases} x \geq 0 \\ x - y < 2 \\ xy \leq 2 \end{cases}$$

Question 7 (18 marks)

(a) For the parabola $y = 2x^2 + 4x + 3$,

(i) find the equation of the axis of symmetry. (1)

(ii) find the vertex. (2)

(iii) sketch the parabola, labelling the vertex, y intercept and axis of symmetry. (3)

(iv) state the minimum value of $2x^2 + 4x + 3$ (1)

(b) Solve $12m - 4m^2 > 0$ (2)

(c) Find where the curve $y = 2x^2 + 5x - 12$ cuts the x axis. (2)

(d) Solve the equation $3x^2 = 2x + 2$ writing your answers in simplest exact form. (3)

(e) Find the point(s) of intersection of the line $y = 6 - 3x$ and the curve $xy = 3$ (3)

(f) The quadratic equation $x^2 - 2x + k = 0$ has roots $x = 1 \pm \sqrt{2}$. Find k . (1)

SOLUTIONS

Question 1

a) $1.6 : 0.12 = 160 : 12$
 $= 40 : 3$

b) $0.00618 = 6.18 \times 10^{-3}$

c) $\frac{43}{56} \times 100\% = 76.78\%$
 $= 77\%$

d) $270 \text{ mm} = 0.27 \text{ m}$

e) $\sqrt[5]{710,000} = (710,000)^{\frac{1}{5}}$
 $= 14.799$
 $= 15 \text{ to 2 sign. figs}$

f) $-2 \leq x+1 < 3$
 $-3 \leq x < 2$

$\therefore x = \{-3, -2, -1, 0, 1\}$

g) $\frac{\sqrt{2}}{2-\sqrt{2}} = \frac{\sqrt{2}}{(2-\sqrt{2})(2+\sqrt{2})}$
 $= \frac{2\sqrt{2}+2}{4-2}$
 $= \frac{2(\sqrt{2}+1)}{2}$
 $= \sqrt{2}+1$

Question 2

a) $100 - 30(20-t) = 100 - 600 + 30t$
 $= 30t - 500$

b) $(x-3y)^2 = x^2 - 6xy + 9y^2$

c) $64 - 8a^3 = 8(8 - a^3)$
 $= 8(2-a)(4+2a+a^2)$

d) $\frac{6-x}{2} \leq x$

$6-x \leq 2x$

$6 \leq 3x$

$2 \leq x$

$x \geq 2$

e) $\left(\frac{1}{27}\right)^{x-2} = (\sqrt{3})^{2x}$

$\left(\frac{1}{3^3}\right)^{x-2} = \left(3^{\frac{1}{2}}\right)^{2x}$

$(3^{-3})^{x-2} = 3^x$

$3^{6-3x} = 3^x$

$6-3x = x$

$6 = 4x$

$x = \frac{3}{2}$

f) $3x-4 = 2x-11 \neq -(3x-4) = 2x-11$

$x = -7$

$-3x+4 = 2x-11$

Which does not satisfy

$15 = 5x$

$3 = x$

Which satisfies

$\therefore x = 3 \text{ only.}$

Question 3

a) $BD = \sqrt{(3+12)^2 + (1+4)^2}$
 $= \sqrt{25^2 + 5^2}$
 $= \sqrt{625 + 25}$
 $= \sqrt{650}$

b) Midpoint of AC = $\left(\frac{-3+4}{2}, \frac{7-10}{2}\right)$
 $= \left(\frac{1}{2}, -\frac{1}{2}\right)$

Midpoint of BD = $\left(\frac{13-12}{2}, \frac{1-4}{2}\right)$
 $= \left(\frac{1}{2}, -\frac{1}{2}\right)$

= Midpoint of AC

$\therefore AC \neq BD$ bisect each other

c) $M_{AC} = \frac{-10-7}{4+3} = \frac{-17}{7}$
 $M_{BD} = \frac{-4-1}{-12-13} = \frac{-5}{-25} = \frac{1}{5}$

$M_{AC} \times M_{BD} = \frac{-17}{7} \times \frac{1}{5}$
 $= \frac{-17}{35}$

$\neq -1$

$\therefore AC$ is NOT perpendicular to BD

d) ABCD is a parallelogram since the diagonals AC & BD bisect each other

e) Equation of BD is
 $y - (-4) = \frac{1}{5}(x - (-12))$
 $5(y+4) = x+12$
 $5y+20 = x+12$
 $x-5y-8 = 0$

f) $d = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$
 $= \left| \frac{1(-3) + (-5)(7) + (-8)}{\sqrt{1^2 + (-5)^2}} \right|$
 $= \left| \frac{-3 - 35 - 8}{\sqrt{1+25}} \right|$
 $= \left| \frac{-46}{\sqrt{26}} \right|$
 $= \frac{46}{\sqrt{26}}$

Question 4

a) Third angle in triangle is 45°

$$\frac{x}{\sin 60^\circ} = \frac{8}{\sin 45^\circ}$$

$$x = \frac{8 \sin 60^\circ}{\sin 45^\circ}$$

$$= 8 \times \frac{\sqrt{3}}{2}$$

$$= \frac{1}{\sqrt{2}}$$

$$= 4\sqrt{3} \times \frac{\sqrt{2}}{1}$$

$$= 4\sqrt{6}$$

b) $\cos \theta = \frac{7^2 + 12^2 - 15^2}{2 \times 7 \times 12}$

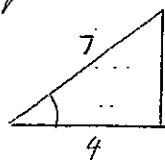
$$= \frac{49 + 144 - 225}{168}$$

$$= \frac{-32}{168}$$

$$= \frac{-4}{21}$$

c) $\cos \alpha = \frac{4}{7}$ & $\sin \alpha < 0$

$\therefore \alpha$ lies in the 4th quadrant.



Pythagoras Theorem

$$\therefore \tan \alpha = -\frac{\sqrt{33}}{4}$$

d) $\sin x = \sqrt{3} \cos x$

$$\frac{\sin x}{\cos x} = \sqrt{3} \quad \text{See note below}$$

$$\tan x = \sqrt{3}$$

$$x = 60^\circ, 240^\circ$$

Note: Check if $\cos x = 0$
 $x = 90^\circ, 270^\circ$

But these two values do not satisfy the given equation.

e) L.H.S. = $(\operatorname{cosec}^2 x - 1)(1 - \cos^2 x)$

$$= \cot^2 x \times \sin^2 x$$

$$= \frac{\cos^2 x}{\sin^2 x} \times \sin^2 x$$

$$= \cos^2 x$$

$$= \text{R.H.S.}$$

Question 5

a) $\hat{BCD} = \hat{ADE}$ (corresp. $\angle$'s, $AD \parallel BC$)

$$= 65^\circ$$

$\hat{BDC} = \hat{BCD}$ (equal $\angle$'s opposite equal sides of isosceles Δ)

$$= 65^\circ$$

$\hat{CBD} = 180^\circ - (\hat{BCD} + \hat{BDC})$
 (Angle sum of ΔBCD)

$$= 180^\circ - (65^\circ + 65^\circ)$$

$$= 50^\circ$$

b) (i) Proof: $\hat{R}$ is common to both Δ 's

$$\frac{PR}{RQ} = \frac{36}{30} = \frac{6}{5}$$

$$\frac{RX}{RY} = \frac{24}{20} = \frac{6}{5}$$

$\therefore \Delta PRO \parallel \Delta XRY$

(one pair of corresponding sides are in proportion and an equal included angle)

(ii) $\hat{RPQ} = \hat{RXY}$
 (corresp. $\angle$'s of similar Δ 's PRO and XRY)

(iii) $PQ \parallel XY$ since corresponding angles RPQ and RXY are equal.

(iv) $\frac{PQ}{XY} = \frac{PR}{XR}$

(corresp. sides of similar Δ 's are in proportion)

$$\frac{PQ}{16} = \frac{3}{2}$$

$$PQ = \frac{3}{2} \times 16$$

$$PQ = 24 \text{ cm}$$

c) Sum of Interior $\angle$'s = $(n-2) \times 180^\circ$

$$3780^\circ = (n-2) \times 180^\circ$$

$$21 = n-2$$

$$n = 23$$

$\therefore$ Polygon has 23 sides.

d) i) $\hat{ADX} = \hat{CBY}$
 (opposite $\angle$'s of a parallelogram are equal)

(ii) $AD = BC$
 (opposite sides of a parallelogram are equal)

$BC = CY = AX$ (given)
 $AD = AX$ (equal to equals)

(iii) Proof $\hat{AXD} = \hat{ADX}$
 (equal $\angle$'s opp. equal sides of isos. Δ)

$\hat{C'D} = \hat{CBY}$
 (equal $\angle$'s opp. equal sides of isos. Δ)

$$\therefore \hat{AXD} = \hat{B'Y}$$

(equal to equal $\angle$'s)

$$\hat{ADX} = \hat{CBY}$$

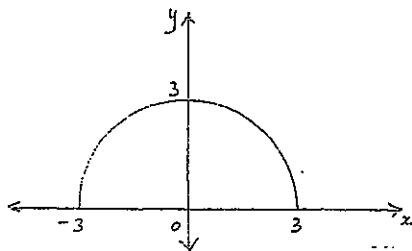
(proven in (i))

$AD = BC$
 (equal sides of parallelogram)

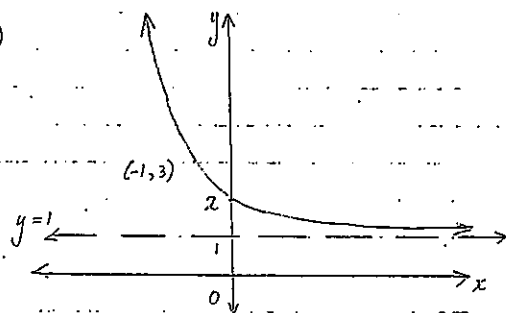
$$\angle ANX = \angle CRX \quad (\text{A.A.C.})$$

QUESTION 6

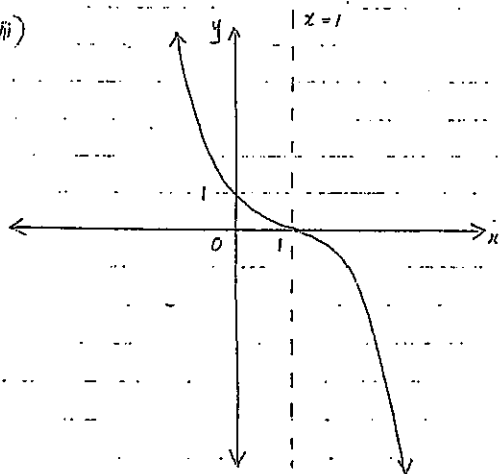
(i)



ii)



iii)



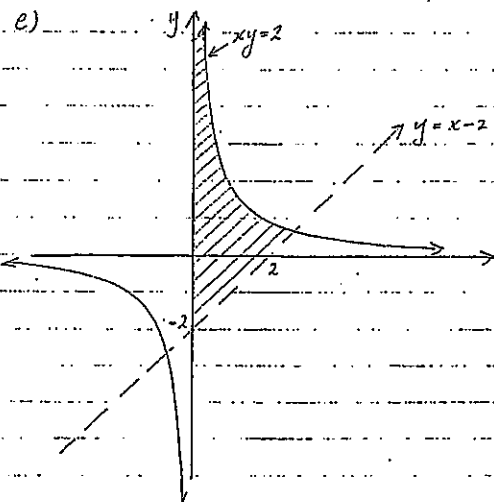
- b) Domain: All real x but $x \neq 2$
 Range: All real y but $y \neq 0$

c) $f(-1) = (-1)^2 + 6 = 7$
 $f(-12) = 6$
 $f(5) = 3 - 5 = -2$

$$\therefore f(-1) + f(-12) - f(5) = 7 + 6 - (-2) = 15$$

$$\begin{aligned} \text{d) } g(-x) &= \frac{3(-x)}{3 + (-x)^2} \\ &= \frac{-3x}{3 + x^2} \\ &= -g(x) \end{aligned}$$

$\therefore g(x)$ is ODD



For $x - y \leq 2$ Test (0,0)
 $0 \leq 2$ True

For $xy \leq 2$ Test (0,0)
 $0 \leq 2$ True

QUESTION 7

a) $y = 2x^2 + 4x + 3$

i) $x = \frac{-b}{2a}$

$$x = \frac{-4}{4}$$

$x = -1$ is Axis of Symmetry

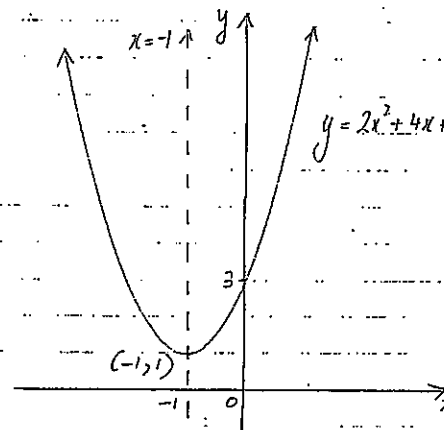
ii) when $x = -1$,

$$y = 2(-1)^2 + 4(-1) + 3$$

$$y = 1$$

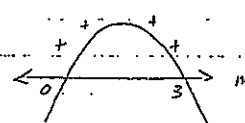
$\therefore$ Vertex is $(-1, 1)$

iii)



iv) Minimum value of $2x^2 + 4x + 3$
 is 1

b) $4m(3-m) > 0$



$$\therefore 0 < m < 3$$

c) Parabola cuts the x axis
 when $y = 0$

$$\therefore 2x^2 + 5x - 12 = 0$$

$$(2x-3)(x+4) = 0$$

$$\therefore x = \frac{3}{2}, -4$$

Parabola cuts the x axis
 at $(\frac{3}{2}, 0) \neq (-4, 0)$

d) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
 $x = \frac{2 \pm \sqrt{4 - 4(3)(-2)}}{2 \times 3}$
 $x = \frac{2 \pm \sqrt{28}}{6}$
 $x = \frac{2 \pm 2\sqrt{7}}{6}$
 $x = \frac{1 \pm \sqrt{7}}{3}$

e) At pt. of intersection

$$x(6-3x) = 3$$

$$6x - 3x^2 = 3$$

$$3x^2 - 6x + 3 = 0$$

$$x^2 - 2x + 1 = 0$$

$$(x-1)^2 = 0$$

$$x = 1, y = 3$$

f) $k = \frac{(1+\sqrt{2})(1-\sqrt{2})}{1-2}$
 $= \frac{1-2}{-1}$
 $= -1$